Derived from: Project_NTI.ipynb & notebook63be3b74d5 (3).ipynb

Hotel Booking Cancellation

An executive data-driven analysis and predictive modeling strategy to forecast, analyze, and mitigate guest cancellations.

Exploration & Engineering

Understanding critical data drivers and engineering features to map booking cancellation behaviors.

Advanced Feature Engineering

Engineered Variables

Derived actionable metrics to capture customer commitment and spatial configurations:

- **Total Guests:** Sum of adults and children.
- **Weekend Ratio:** Percentage of nights spent over weekends.
- **Price per Guest:** Room price scaled dynamically per occupant.

Data Preprocessing Pipeline

Applied rigorous math transformations to prepare unstructured rows for classifier optimization:

- **Log Scaling:** Applied log1p to correct skewed numeric distributions.
- **SMOTE Integration:** Balanced target variables dynamically.
- **MinMax Scaler:** Bound values cleanly between 0 and 1.

Key Exploratory Data Insights

Lead Time Influence

Bookings engineered with long lead times are exponentially more likely to result in a cancellation, as guests find alternative stays.

Special Requests

Guests who include one or more special requests demonstrate far higher reservation commitment, significantly lowering cancellation rates.

Loyalty Impact

Repeated guests are incredibly stable customers, showing near-zero cancellation risk across all analyzed months.

Predictive Architecture

Evaluating standard machine learning frameworks against feedforward neural networks.

Comparing the Model Architectures

Random Forest Classifier

A non-parametric ensemble method optimized using 200 trees, 15 max depth, and robust leaf splitting criteria.

Excels wonderfully in tabular classification, handling sparse categorical dummies smoothly while mapping complex non-linear booking patterns.

Sequential Deep Learning

A 5-layer multilayer perceptron equipped with progressive dropout layers (0.4 to 0.2) to mitigate overfitting.

Compiled via binary cross-entropy and optimized using Adam learning-rate scheduling. Highly expressive but susceptible to slight tabular overfitting.

Model Accuracy & Recall Comparison

As illustrated above, Random Forest consistently outperforms the Sequential Deep Neural Network across both raw test accuracy and critical cancellation recall rates.

Exceptional Cancellation Recovery

76.1%

Cancellation Recall Rate

Unlocking Business Value

Our optimized Random Forest model successfully flags over 76% of all booking cancellations well in advance of the scheduled arrival date.

By identifying high-risk reservations proactively, hotel management can implement strategic pre-payment nudges and secure room revenue.

Detailed Performance Metrics

Model Candidate	Overall Accuracy	Recall (Cancellations)	Precision (Cancellations)	F1-Score (Cancellations)
Random Forest (Optimal)	82.58%	76.05%	67.00%	0.71
Deep Neural Network	79.27%	70.20%	62.00%	0.66

Business Strategies

Translating predictive model results into actionable hotel operational excellence.

Actionable Business Strategies

Dynamic Overbooking Policies

Utilize the Random Forest's 76.1% predictive recall to safely overbook predicted cancellations during peak seasons, optimizing total occupancy without risk.

Targeted Upfront Deposits

Enforce strict pre-payment or non-refundable reservation policies specifically for bookings flagged by the model as high-risk (such as long lead-time reservations).

Incentivize Guest Commitment

Because bookings with special requests and returning customers almost never cancel, provide easy-to-use check-in customization portals to lock in guest commitment early.

Questions & Answers

Strategic predictive analytics for hotel cancellation mitigation.

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